

Pascal's triangle



- 1
- a 1, 5, 10, 10, 5, 1
- b 32
- c ${}^5C_0, {}^5C_1, {}^5C_2, {}^5C_3, {}^5C_4, {}^5C_5$

- 2 1, 8, 28, 56, 70, 56, 28, 8, 1

- 3 7th

- 4
- a 7
- b 9

- 5
- a 5
- b 4

- 6 $(x + m)^4 = 1x^4m^0 + 4x^3m^1 + 6x^2m^2 + 4x^1m^3 + 1x^0m^4$

Binomial theorem

- 7
- a 6
- b 14
- c 10
- d 1

- 8
- a $560b^3$
- b $126c^5d^4$
- c $135u^2v^4$
- d $\frac{55}{128}u^3$
- e $\frac{280}{3y^2}$
- f $2288x^3y^{20}$
- g $-3876x^4y^{75}$
- h $1\,451\,520x^8y^{12}$
- i $216x^{-6}y^{-4}$
- j 1792

- 9 $45x^2y^8, 10xy^9, y^{10}$

- 10
- a $n = 10$
- b $45m^2k^8$

- 11 $p = 2$



13 4C_3

14 $(4p + 3q)^3 = {}^3C_0 (4p)^3 (3q)^0 + {}^3C_1 (4p)^2 (3q)^1 + {}^3C_2 (4p)^1 (3q)^2 + {}^3C_3 (4p)^0 (3q)^3$

15

a $256x^4$

b $81y^4$

16

a $\binom{5}{2}$

b 10

c 4th term

17 $55u^2v^9, 55u^9v^2$

18 10 777 536

19

a $1 + 4x + 6x^2 + 4x^3 + x^4$

b $32 + 80b + 80b^2 + 40b^3 + 10b^4 + b^5$

c $p^5 + 5p^4q + 10p^3q^2 + 10p^2q^3 + 5pq^4 + q^5$

d $c^8 - 8c^7d + 28c^6d^2 - 56c^5d^3 + 70c^4d^4 - 56c^3d^5 + 28c^2d^6 - 8cd^7 + d^8$

e $u^6 + 24u^5v + 240u^4v^2 + 1280u^3v^3 + 3840u^2v^4 + 6144uv^5 + 4096v^6$

f $y^4 + 12y^3 + 54y^2 + 108y + 81$

g $(ka)^6 - 6(ka)^5b + 15(ka)^4b^2 - 20(ka)^3b^3 + 15(ka)^2b^4 - 6kab^5 + b^6$

h $y^3 - 15y^2 + 75y - 125$

i $y^3 - y^2 + \frac{y}{3} - \frac{1}{27}$

j $8 + 6y + \frac{3y^2}{2} + \frac{y^3}{8}$

k $27y^3 + 54ry^2 + 36r^2y + 8r^3$

l $a^3 - 3a + \frac{3}{a} - \frac{1}{a^3}$

m $u^6 + 9u^4v^2 + 27u^2v^4 + 27v^6$

n $1 + \frac{12}{y} + \frac{48}{y^2} + \frac{64}{y^3}$

20

a $405x^4$

b $120c^3d^7$

c $165u^3v^8$

d $3003u^5v^{10}$

21 $n = 16$

$$\frac{21}{2}$$



23 $\binom{9}{3}$

24 9

25 ${}^8C_k \times 3^{8-k} (-x)^k$

26

a 0.943

b 1.037

c 0.890